

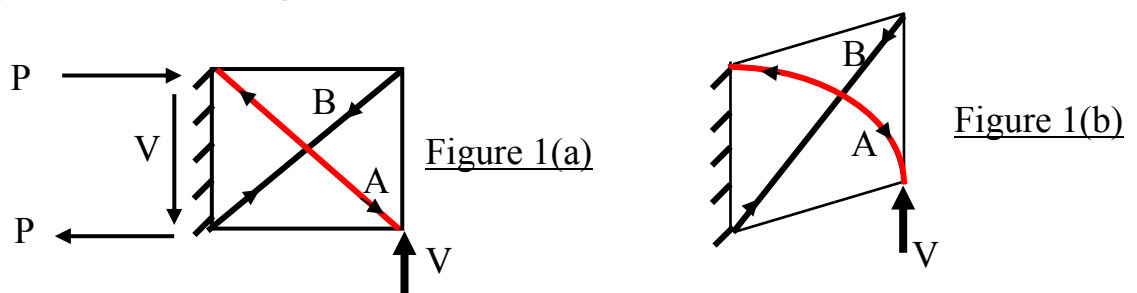
TENSION FIELD SHEAR BEAMS

Types of Shear Beams

The transmission of shear in a beam can be achieved in several ways namely:-

- Diagonal bars
- Shear web-
 - i) Thick-shear resistant
 - ii) Thin - shear buckling

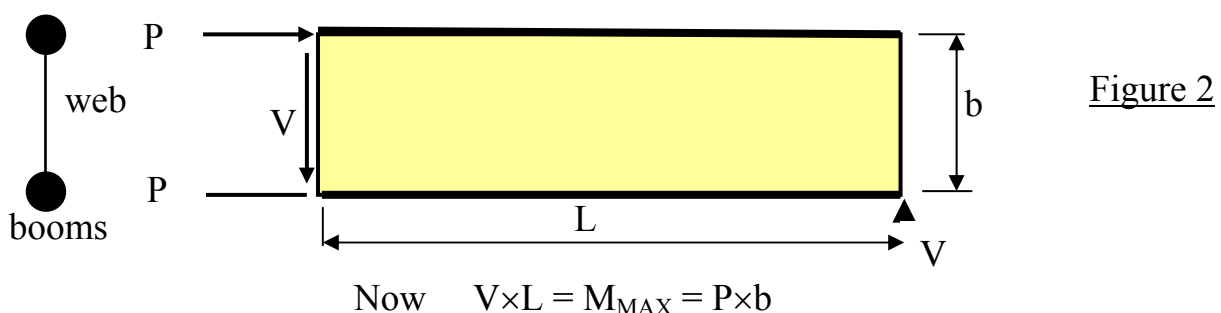
For an open truss with diagonal bars



Load V causes a compressive load in bar A and a tensile load in bar B as shown in Figure 1(a). As the load is increased bar A buckles (Figure 1(b)) and can carry no further increase in load, thus additional increases in V are carried solely by tension in bar B.

For a shear web.

The diagonal members of the truss are replaced by a sheet web (see Figure 2). The section is often idealised such that the web carries the shear and the booms (flanges) carry the bending moment as axial end loads.

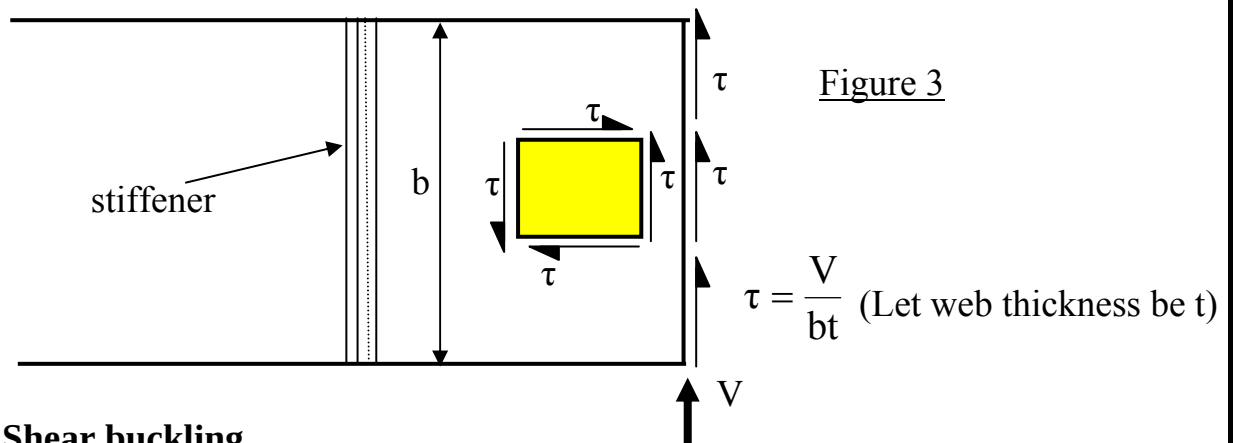


The maximum boom direct stress (due to the bending moment only) is $\sigma = P/A$, where A is the cross section area of each boom.

The shear load is balanced by shear stresses in the web.

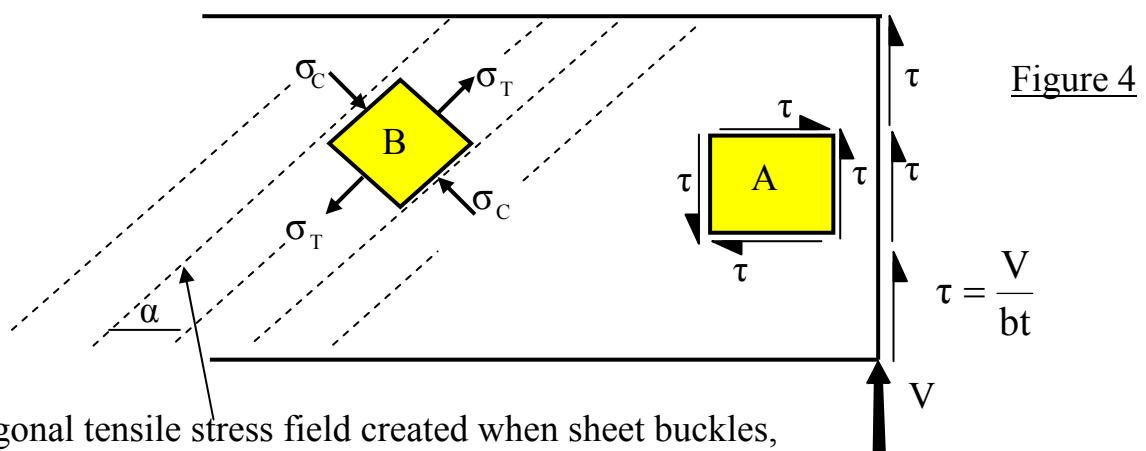
(i) Shear resistant web.

The analogy here is to a short stocky strut which does not fail by buckling. Likewise the web will not fail until the web shear stress $\tau (= V/bt)$ reaches the yield shear stress of the material (see Figure 3).



(ii) Shear buckling.

If the thick web is now replaced by a thin web, having a low resistance to buckling (i.e. compression) then the web exhibits a different failure mode. Consider an element of the web, shown in Figure 4, at the neutral axis (ie bending is absent). (A) shows an element in pure shear but this can be replaced by element (B) at 45 degrees which is subjected to direct stresses only as presented in a previous lecture. In this case $\sigma_c = \sigma_t = \tau$. When the critical compressive stress of the web is reached the web buckles and forms into **diagonal tension fields**. In the limit the compression stress can be taken as equal to zero which means that all the load is being carried in tension by the web which is now acting like the diagonal tie bars in the truss structure.



Diagonal tensile stress field created when sheet buckles, in the limit σ_c considered to be zero. α is approx. 45 degrees.

The magnitude of the tensile forces generated by the tension field gives rise to the following load cases which are illustrated in Figure 5:-

- (i) compressive loads in the stiffeners
- (ii) compressive loads in the flanges
- (iii) bending (sagging) loads in the flanges

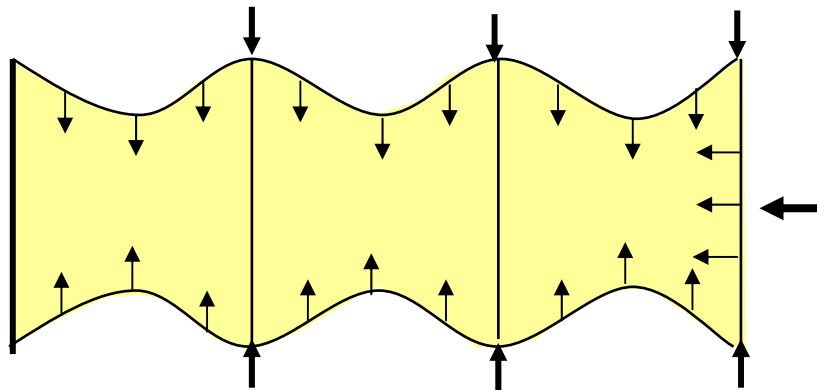


Figure 5

Hence the tensile stress field set up in the thin web of the panel leads to a beam construction similar to that shown in Figure 6, consisting of stringers and flanges which frame the web into separate panels and which are capable of carrying the generated loads which must be determined.

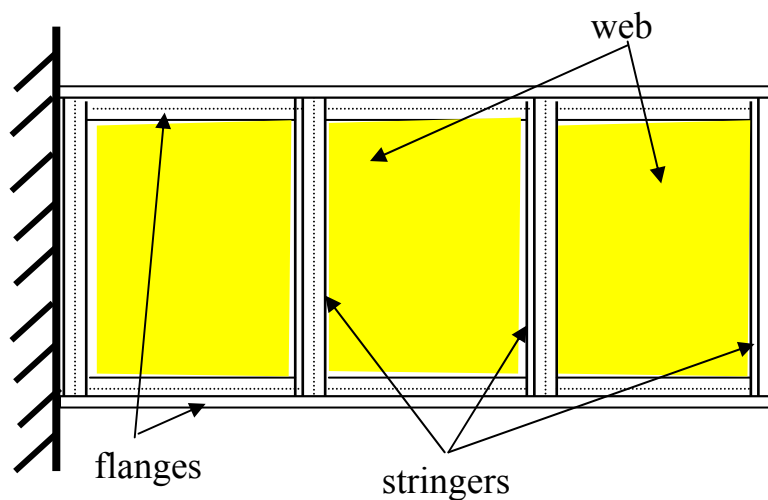


Figure 6 --Typical tension field shear beam construction

Complete Diagonal Tension

Evaluation of tension field stress

The post buckling stresses and forces on a section of the web of a beam subjected to a load V can be represented as shown in Figure 7.

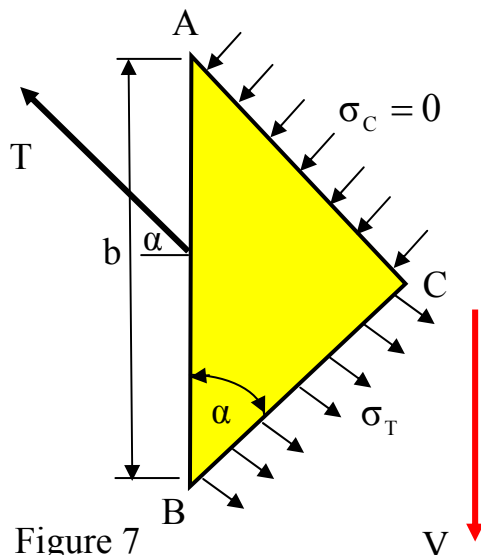


Figure 7

The force on $AC = 0$ (from our initial assumption)

The length $BC = AB \times \cos \alpha = b \cos \alpha$

Then area of the web section $BC = bt \cos \alpha$

$$\therefore T = \sigma_T bt \cos \alpha \quad \text{hence } \sigma_T = T/bt \cos \alpha$$

$$\text{now } T \sin \alpha = V$$

$$\therefore \sigma_T = \frac{V}{bt \sin \alpha \cos \alpha} \quad \text{but } V/bt = \tau$$

$$\sigma_T = \frac{\tau}{\sin \alpha \cos \alpha} = \frac{2\tau}{\sin 2\alpha}$$

For a value of α of 45 degrees then $\sigma_T = 2\tau = \frac{2V}{bt}$ twice the applied shear stress.

Force induced in the vertical stiffeners

When the web is in a state of tension, the tensile forces will be balanced by compressive forces in the stringers of the beam. These forces must be determined to ensure that the stringers do not buckle.

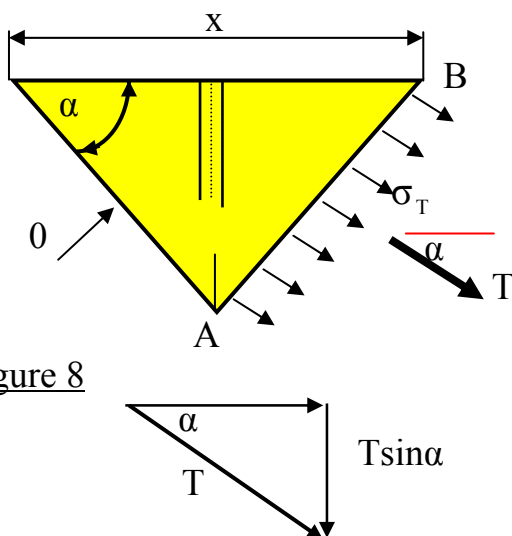


Figure 8

$$\text{Force on } AB = \sigma_T \times t \times AB = T$$

$$\text{Vertical component} = (\sigma_T \times t \times AB) \sin \alpha$$

Let $x = a$ (one stringer pitch) then:-

$$AB = x \sin \alpha = a \sin \alpha$$

Thus the vertical component /stringer, P_s

$$= \left(\frac{\tau \times t \times a \sin \alpha}{\sin \alpha \cos \alpha} \right) \sin \alpha \quad \text{but } \tau = V/bt \text{ so}$$

$$P_s = \frac{V \times t \times a \sin \alpha \sin \alpha}{bt \times \sin \alpha \cos \alpha} = \frac{V \times a \times \tan \alpha}{b}$$

Force induced into flanges

The tensile forces are balanced in the longitudinal direction of the beam by compressive forces in the beam flanges as shown in figure 9

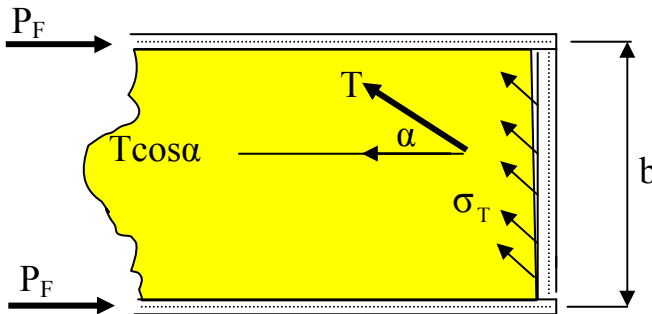


Figure 9

Horizontal force on end stringer is

$$T \cos \alpha = \sigma_T (b \times t \times \cos \alpha) \cos \alpha$$

and substituting for σ_T

$$= \frac{V \times b \times t \times \cos \alpha \cos \alpha}{b \times t \times \sin \alpha \cos \alpha} = V \cot \alpha$$

therefore additional force per flange is:-

$$P_F = \frac{V \cot \alpha}{2}$$

Sagging bending of flanges.

The component of the web tension normal to the flanges (used to determine the load in the stringers) also causes secondary bending of the flanges. Since this effect tends to cause the flanges to “sag” between stiffeners, it is often called sagging bending. Assuming that the loading causing this sagging bending is uniformly distributed along the flanges and taking the flanges as being supported (built-in) at the stiffeners then the maximum bending moment on a flange is at a stiffener and being $w a^2/12$ where w is the load per unit length.

$$\text{Thus } w = \frac{V \times a \times \tan \alpha}{b} \div a = \frac{V \tan \alpha}{b}$$

$$\text{Therefore maximum moment on flange} = \frac{V a^2 \tan \alpha}{12b}$$

Maximum stress in the flanges is thus:- $\sigma = \pm \frac{M}{bA} - \frac{V \cot \alpha}{2} \times \frac{1}{A} \pm \frac{V a^2 \tan \alpha}{12b} \times \frac{y}{I}$

$$\left(\pm P_A\right) \quad \left(-P_F/A\right) \quad \left(\pm M_{sy}/I\right)$$

where y is maximum distance of the flange section from flange neutral axis, I is the second moment of area of flange section and A is the flange cross section area. M is the maximum primary bending moment.

Buckling of Flat Plates in Shear.

The web buckling equation has a similar form to that for the compressive buckling stress of plates which was derived in an earlier lecture namely:-

$$\tau_b = \left[\frac{\pi^2}{12(1-\nu^2)} \right] K_s E \left(\frac{t}{b} \right)^2$$

Note that in this equation b is the shorter dimension of the panel dimensions (a and b) shown in Figure 10. K_s is obtained from Figure 11 and $\pi^2/12(1-\nu^2)$ is taken as 0.9 for a Poissons ratio of 0.3.

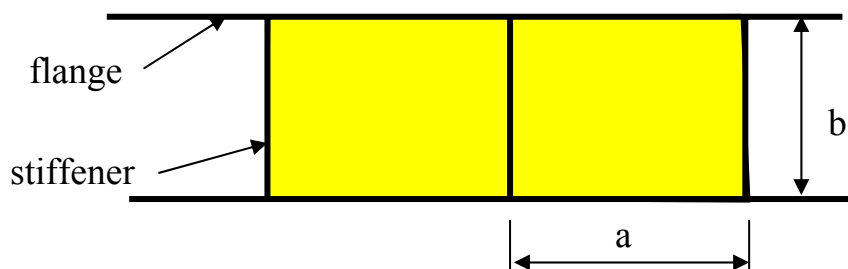
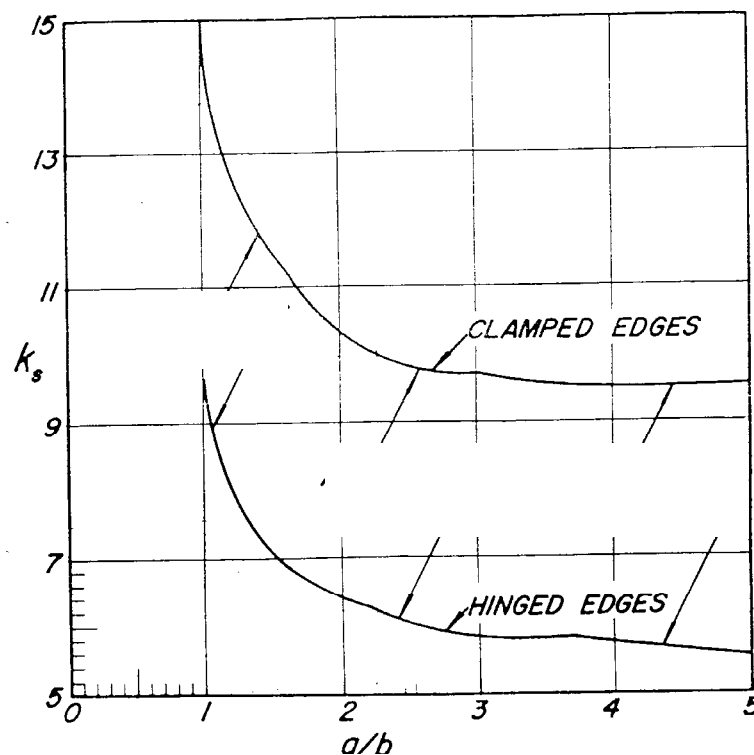


Figure 10



Shear Buckling Stress Coefficient Figure 11

Example 1

A cantilever beam, shown in Figure 12, is idealised so that the web carries only shear and the flanges direct stress.

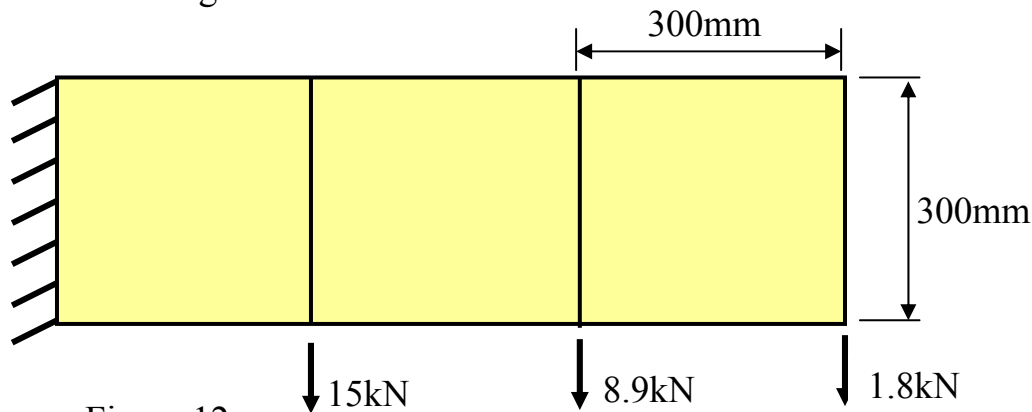


Figure 12

Calculate the web thickness for the shear beam for:-

- buckling as the criteria
 - treating the web as shear resistant (i.e. yield stress as the failure mode)
- Shear yield stress = 85 MPa $E = 70\text{kN/mm}^2$ $\nu = 0.3$

For buckling as the criteria

$a/b = 300/300 = 1$ hence from Figure 11 $K_s = 9.5$ (assuming hinged edges)

$$\tau_b = K_s E \left(\frac{t}{b} \right)^2 = 0.9 \times 9.5 \times 70 \times 10^3 \times \left(\frac{t}{300} \right)^2$$

$$\tau_b = 6.678t^2 \dots\dots\text{equ.1}$$

Shear Diagram for the beam

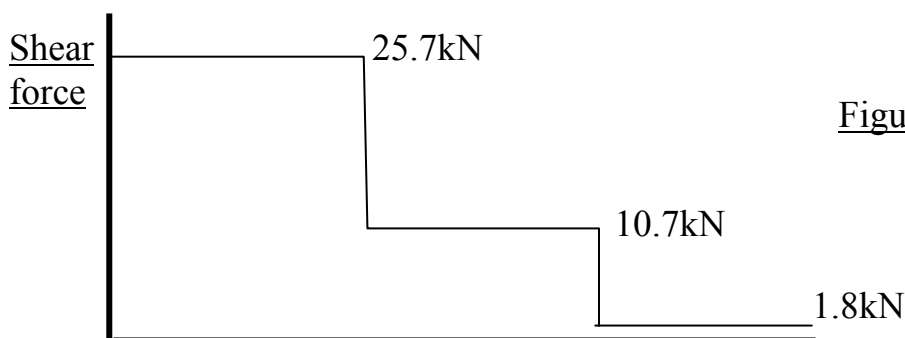


Figure 13

Cross section area of skin = $300 \times t$

Maximum shear load in a panel = 25.7 kN

$$\tau = \frac{25.7 \times 10^3}{300t} \dots \text{equ.2}$$

Equating (1) and (2) then $\frac{25700}{300t} = 6.678t^2$

$$\text{therefore } t = \sqrt[3]{12.83} = \underline{2.34 \text{ mm}}$$

The web would need to be 2.34 mm for the buckling threshold to occur at 25,700 N.

For a shear resistant design.

Shear stress = yield stress

$$\frac{25700}{300t} = 85 \quad t = \underline{1.07 \text{ mm}}$$

Hence if buckling was ignored in the design and the thickness of the web designed on shear resistance only, the strength analysis would have been incorrect.

See tutorial questions 1 and 2

If it is assumed that the flanges and stiffeners of the beam are rigid then $\alpha = 45^\circ$. In practice however, both flanges and stiffeners deform so that α is less than 45° , usually of the order of 40° . The angle adjusts itself so that the total strain energy of the beam is a minimum.

There are a number of expressions for α , one being $\tan^4 \alpha = \frac{1 + tb/2A_F}{1 + ta/A_S}$

where A_F is the cross-sectional area of each flange and A_S is the cross-sectional area of a stiffener.

If the load on a stiffener is high enough then it will buckle. Tests indicate that stiffeners will buckle as columns of equivalent length as follows:-

$$L_e = b/(4-2a/b)^{0.5} \quad \text{for } a < 1.5b$$

$$L_e = b \quad \text{for } a > 1.5b$$

In addition the other stiffener failure modes considered in the section “compression panels” must be addressed.

A stringer is fully effective in resisting the load if the centroid of the stringer lies in the plane of the beam web, ie when the stringer consists of two members symmetrically arranged on opposite sides of the web. Where the web is stiffened by a single member attached to one side of the web only, then the compressive load is offset from the stringer axis thus producing a bending load in addition to the axial load and eccentric load formula must be used.

Incomplete diagonal tension is the case where the web is significantly thick so that the compressive buckling stress of the web should be considered in the beam analysis to obtain a lightweight design.

It is assumed that the applied web shear τ may be divided into a shear resistant component τ_S and a diagonal tension component τ_{DT} by writing

$$\tau_{DT} = k\tau \quad \text{and} \quad \tau_S = (1 - k)\tau$$

where k is known as the *diagonal tension factor* and is a measure to which the diagonal tension is developed. A completely unbuckled web has $k = 0$ and $k = 1$ for a web in complete diagonal tension. A modified set of equations involving k and empirical formula are used to apply this approach.

The value of k for a web with a buckling shear stress of τ_{CR} is given by the empirical expression

$$k = \tanh\left(0.5 \times \log \frac{\tau}{\tau_{CR}}\right)$$

the ratio τ/τ_{CR} is known as the *loading ratio*.

The buckling stress is calculated from the formula:-

$$\tau_{CR} = k_{ss} E \left(\frac{t}{b}\right)^2 \left[R_b + \frac{1}{2} (R_a - R_b) \left(\frac{a}{b}\right)^3 \right]$$

where k_{ss} is the plate coefficient for simply supported edges and R_a and R_b are empirical restraint coefficients (ref. Bruhn) for the vertical and horizontal edges of the web panel.

The flange and stiffener stress equations are modified to:-

$$\sigma_F = \frac{k \tau \cot \alpha}{(2A_F/tb) + 0.5(1-k)} \quad \text{and} \quad \sigma_S = \frac{k \tau \tan \alpha}{(A_S/ta) + 0.5(1-k)}$$

The flange and stiffener flexibility cause the web stress in the area around the stringer to increase and to decrease between the stringers thus reducing the flange bending moment and hence the sagging stresses (ref. Bruhn for coefficients).

Tutorial

Question 1

A shear panel cantilever has a web depth of 300 mm. The cantilever is subjected to a shear force of 20kN causing a fully developed diagonal tension field in which the tensile stress is 70 N/mm^2 (Take $\alpha = 45^\circ$).

Calculate:-

- The cross sectional area of the skin
- The stringer spacing if the compressive force in the stringers is 25kN.
- The additional force in each flange
- The buckling stress of the web ($\nu = 0.3$)

$$E = 65,000 \text{ N/mm}^2$$

Assume hinged edges.

Question 2

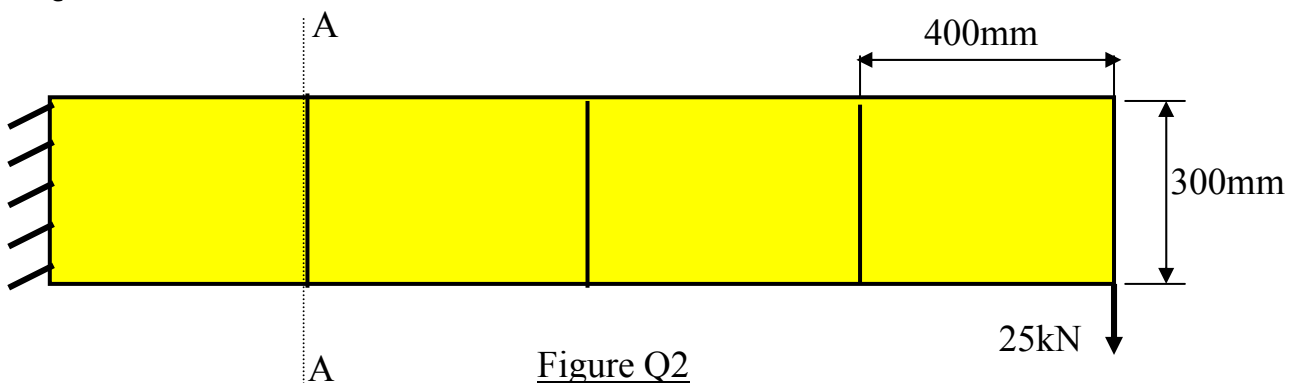


Figure Q2 shows a four panelled shear beam. Assuming a fully developed diagonal tension field in the web, find for the cantilever beam :-

- The tensile stress in the web
- The compressive stress in the stringers
- The maximum compressive stress in the flange at section A-A. Take I for the flange as $30,000 \text{ mm}^4$ and the maximum distance from the neutral axis to the outermost fibres as 15 mm.
- The web buckling stress (assume hinged edges)

Other data:

Web thickness = 2 mm

Cross section of stringers = 200 mm^2

Assume $\alpha = 45^\circ$ $E = 70,000 \text{ N/mm}^2$

Cross section of flange = 1000 mm^2